5.4 Digital currency

When you pay for products and services using the internet, you will normally use a **digital currency**. A digital currency is one that exists electronically (see Figure 5.16). It is a method of payment that is similar to paying with coins and bank notes. However, rather than exchanging physical bank notes and coins to make the payment, the currency is exchanged digitally using computers. There are several different forms of digital currency, the most popular are payments using credit cards, mobile phones and smart watches. The most common of these is using credit cards when making payments using the internet.

When the data about the payment with the digital currency is sent from one computer to another, it is encrypted to keep it secure. The method of encryption used is what you learnt about previously, in Section 5.2, using the HTTPS protocol. The payment details for the use of a digital currency can be stored in a persistent cookie. That means that each time you visit the website to buy a product, your payment details will be automatically entered so you don't need to type them in each time.

KEY WORD

digital currency: a currency that exists electronically rather than physically.



Figure 5.16: Digital currency exists electronically and can be sent between computers

One type of digital currency that you may have heard of is Bitcoin. This type of digital currency is more specifically known as a **cryptocurrency**. The banks that we use to manage our money on a daily basis are called centralised systems. This means that there is an authority at the centre managing the process in which they are used. In this case, the bank is the authority. A cryptocurrency is a decentralised system, this means that there is no central authority, like a bank, managing the process of payments. Payments are also encrypted using this type of currency. As there is no central authority managing the system of payments with this type of digital currency, a different system called **blockchain** is used to keep track of the payments.



Figure 5.17: A cryptocurrency is a digital currency that is managed by a decentralised system

KEY WORDS

cryptocurrency: a type of digital currency that uses encryption procedures.

blockchain: a method that is used to track all the transactions made with a cryptocurrency.

digital ledger: a database that is a list of all the transactions recorded by the use of blockchain.

In its most basic form, blockchain is a list of all the records made with the digital currency. This is called a **digital ledger** (Figure 5.18). Each time a payment is made with the digital currency, a record is added to the ledger that includes a digital signature with the time and date that the payment is made. In blockchaining, once these records are added to the digital ledger, they cannot be changed. Data that is stored in a digital

ledger is also encrypted. This prevents anyone being able to change the data. However, anyone who wants to view the data in the digital ledger is able to do so.

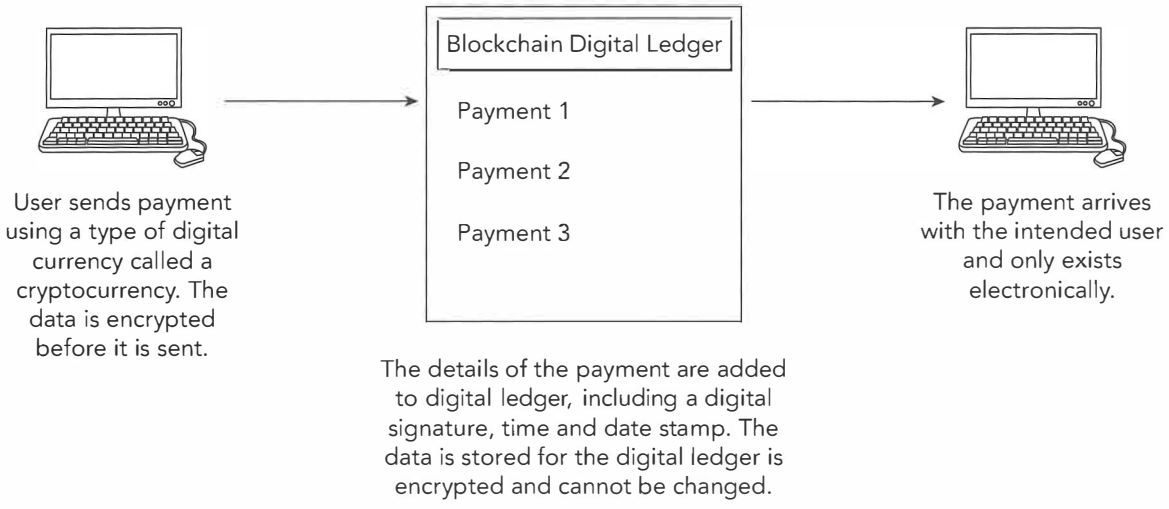


Figure 5.18: A blockchain keeps a digital ledger of transactions

COMPUTER SCIENCE IN CONTEXT

There are as many as 2000 different digital currencies that exist. Many of these are different types of cryptocurrency. How many do you think you could name?

5.5 Cyber security

Whilst the internet and the world wide web can be a very fun and interesting place, it does also have some issues and threats that you definitely need to be aware of. You need to know how these cyber threats are created, what the motivation is behind those who use them, and what kind of security solution you can put in place to try and prevent them from happening. One thing to note is that each security solution may not be a complete prevention, sometimes it is better to have several solutions in place. Also, **perpetrators** are always looking to develop ways to beat security solutions, so no solution will be 100% effective in preventing an attack. However, if you don't have any security solutions in place, you are putting your personal data at a far greater risk of these cyber threats. So, what are the cyber threats that you need to know?

KEY WORDS

perpetrator: a person that carries out an illegal or immoral action.

brute-force attack: a type of cyber threat that involves repeatedly inputting a password until the correct one is found, to try and break into an account or device.

Brute-force attack

You may have heard someone being described as a brute person. This normally means people think they are strong and forceful. This is exactly what a **brute-force attack** is on your computer, a strong and forceful attack. Imagine that someone is banging on a locked door, pushing and shoving it in every way that they can, until the door gives way and opens. In a brute-force attack, a perpetrator tries to break into your computer or account. They do this by repeatedly entering different password combinations until they manage to enter the correct one. They may try to do this manually, by typing in each password combination themselves. They could also do this by developing a